

Homework 1: Getting the tools working

Stefan Fahmy · Algory QI Training

Setup

Setup check output:

ok 5 days pulled

GitHub repository: <https://github.com/stefanfahmy2/algory-qi-f26>

Data: NKE (Nike) vs SPY

I chose Nike (NKE). Prices are one year of daily closes (dividend-adjusted, yfinance default), rows with missing values dropped. Output of HW1.py:

Tickers: NKE, SPY
Trading days: 251
First date: 2025-09-19
Last date: 2026-09-18

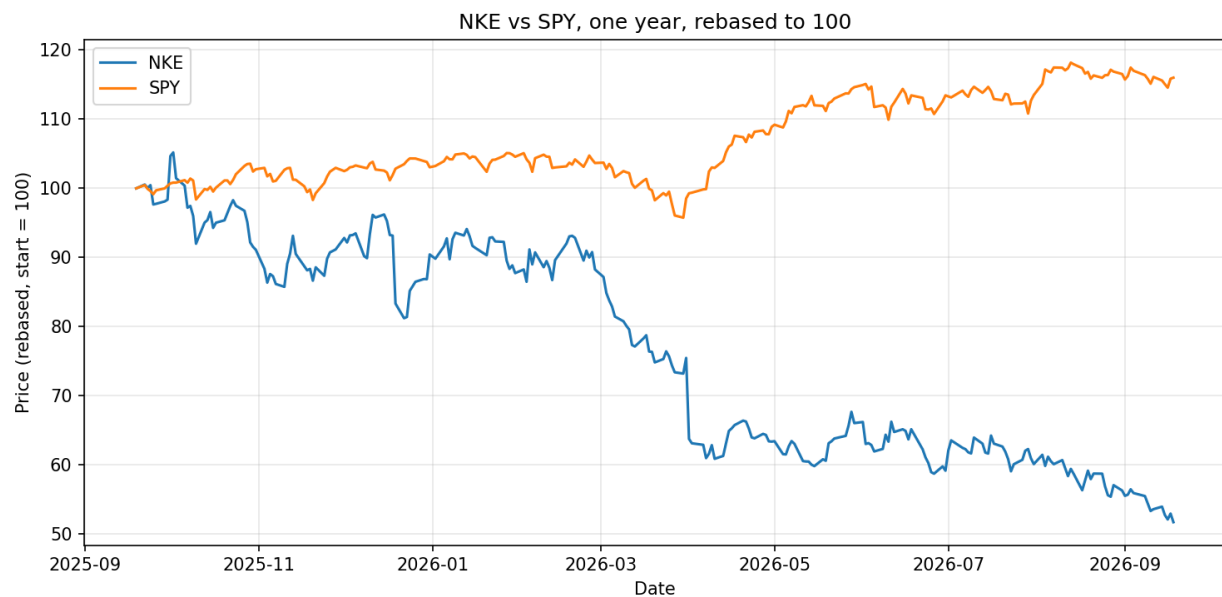
NKE

Last close: 35.51
Return over year: -48.26%
Annualised vol: 36.37%

SPY

Last close: 761.69
Return over year: 16.00%
Annualised vol: 12.93%

Biggest single-day move for NKE: 2026-04-01 (-15.51%)



Biggest single-day move

2026-04-01: -15.51%. Nike fell about 15.5% (closing near \$44.62) the day after its fiscal Q3 report: although earnings beat estimates, management guided to declining revenue over the next few quarters, said China sales would keep falling, and pushed the margin recovery back to fiscal Q2 2027. Analysts responded with downgrades and lower price targets.

What the chart shows

NKE badly lagged the market: it fell about 48% over the year while SPY rose 16%, and much of the damage came in steps (notably the April 1 guidance drop) rather than gradually. It was also a much rougher ride, with annualised volatility of 36% versus SPY's 13%. I would not hold it right now, because I would be taking nearly three times the volatility for a negative return while the turnaround keeps slipping; I would only revisit it once China sales and margins show real recovery.

Career

LinkedIn: completed.

Summer 2027 quant internship applications at three firms (all are rolling, with no fixed deadline, and all are already open):

- **Citadel Securities:** summer 2027 postings live by about July 2026.
- **Optiver** (Chicago quantitative intern): posting appeared around 1 July 2026.
- **Jane Street:** summer 2027 applications reported open from about August 2026 (London trading opened around 3 August; each office runs its own calendar).

Dates are approximate, taken from recruiting trackers; the firms do not publish fixed dates.